

# Vishal Talsaniya

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## Professional Summary

Backend Developer specializing in VoIP technologies and scalable API development. Extensive hands-on experience with FreeSWITCH, OpenSIPS, FusionPBX, and WebRTC for building enterprise telephony solutions. Proficient in developing production-grade RESTful APIs using Go (Golang) with frameworks such as Echo and Beego ORM. Strong expertise in SIP/RTP protocols, Lua scripting, and Docker containerization. Experienced in implementing real-time communication systems with message queuing (RabbitMQ) and managing both SQL (MySQL) and NoSQL (MongoDB) databases. Passionate about building robust VoIP infrastructure and solving complex real-time communication challenges.

## Technical Skills

### VoIP Technologies & Protocols

- FreeSWITCH: SIP call handling, internal/external profiles, RTP media flow, ESL (Event Socket Library), call event handling (CHANNEL\_CREATE, ANSWER, BRIDGE, HANGUP, DESTROY)
- FusionPBX: PBX configuration, extension management, dialplan customization
- OpenSIPS: SIP routing, load balancing, proxy configuration, dispatcher module
- RTPEngine: Media proxying, NAT traversal, codec handling
- SIP Protocol: Call signaling, registration, session management, call setup/teardown
- RTP/RTCP: Real-time media transmission, quality monitoring, jitter and packet loss analysis
- WebRTC: Browser-based real-time communication, SRTP, WebSocket signaling, ICE/STUN/TURN (basic)
- Jitsi: Architecture (Prosody, Jicofo, JVB), SIP-to-WebRTC calling, conferencing

### Programming & Scripting

- Go (Golang): RESTful API development, Echo framework, Beego ORM, microservices
- Lua: FreeSWITCH dialplan scripting, dynamic call routing, call metadata capture, TTS/STT integrations
- JavaScript/Node.js: Basic API development with Express, FreeSWITCH ESL integration (modesl)

### Databases & Infrastructure

- Databases: MySQL, MongoDB, MariaDB
- Message Queue: RabbitMQ (producer/consumer model, Paotia library)
- DevOps: Docker containerization, multi-container orchestration
- Tools: Postman (API testing), Git (GitHub/GitLab)
- Operating Systems: Linux (Ubuntu), Shell scripting

### Network & Transport Protocols

- UDP/TCP: SIP signaling transport, RTP media delivery, WebSocket (WS/WSS) for WebRTC

## Work Experience

### Backend Developer

**Siprtc.io**

*December 2024 – June 2025*

- Developed backend services for VoIP-based applications
- Worked with SIP protocol and real-time communication systems
- Integrated APIs for call handling and notifications
- Troubleshoot and optimized server performance

### Software Engineer (VoIP / FreeSWITCH)

**Vindaloo Softtech Pvt. Ltd.**

*July 2025 – Present*

- Working with **FreeSWITCH** for VoIP call handling and routing
- Configuring SIP profiles, dialplans (XML), and call flows
- Handling NAT, SIP signaling, and media (RTP) issues
- Integrating push notifications with OpenSIPS / VoIP systems
- Debugging call issues and improving system performance

## Professional Experience

### Backend Development

- Developed production-grade RESTful APIs using Golang (Echo framework, Beego ORM) for cloud-based call center platforms
- Implemented VoIP telephony solutions using FreeSWITCH and FusionPBX for enterprise-level communication systems
- Configured and deployed OpenSIPS for SIP load balancing and proxy routing with RTPEngine media handling
- Created custom Lua scripts for FreeSWITCH dialplan logic, dynamic call routing, and event-driven call handling
- Integrated RabbitMQ message queuing (Paota library) for real-time campaign processing and task distribution
- Designed MySQL and MongoDB database schemas for scalable data storage and efficient query performance
- Containerized VoIP applications and microservices using Docker for consistent deployment environments
- Debugged SIP signaling issues, RTP media problems, and ESL connection challenges in production environments
- Conducted comprehensive API testing and documentation using Postman and participated in agile development practices

## VoIP Projects

### SIP Server with OpenSIPS, WebRTC & RTPEngine

*Technologies:* OpenSIPS, RTPEngine, WebRTC, Docker, MariaDB, WebSocket [GitHub Repository](#)

- Built complete SIP server infrastructure using OpenSIPS with WebRTC support for browser-based calling
- Configured RTPEngine for media proxying, NAT traversal, and WebRTC-to-SIP media bridging
- Implemented WebSocket transport for SIP signaling and SRTP for secure media transmission
- Containerized entire stack using Docker for simplified deployment and scalability
- Integrated MariaDB for SIP user authentication and call detail records (CDR)

### VoIP Load Balancer with FreeSWITCH

*Technologies:* FreeSWITCH, OpenSIPS, RTPEngine, Docker, MariaDB, Lua, Google STT API [GitHub Repository](#)

- Designed VoIP load balancer using OpenSIPS dispatcher module to distribute calls across multiple FreeSWITCH instances
- Developed custom Lua dialplans in FreeSWITCH for intelligent call routing and IVR functionality
- Integrated Google Speech-to-Text API for voice recognition and real-time transcription capabilities
- Implemented RTPEngine for high-availability media handling and codec transcoding
- Containerized multi-component VoIP stack using Docker Compose for orchestration

### OpenSIPS Control Panel with Docker

*Technologies:* OpenSIPS, Control Panel, Docker, MySQL, Apache

[GitHub Repository](#)

- Deployed OpenSIPS Control Panel for web-based management of SIP users, domains, and routing rules
- Containerized OpenSIPS CP with Apache web server and MySQL backend using Docker
- Configured user authentication, call routing management, and real-time monitoring dashboards

### VoIP API Development

*Technologies:* Golang, FreeSWITCH ESL, RESTful API, VoIP Integration [GitHub Repository](#)

- Developed RESTful APIs in Golang for VoIP service management and call control operations
- Built endpoints for call origination, call termination, and real-time call status monitoring
- Integrated FreeSWITCH Event Socket Library (ESL) for programmatic call control
- Designed modular architecture for scalable VoIP application development and third-party integrations

### Call Center Service APIs (Production)

*Role:* Backend Developer | Production-Level APIs at siprtc.io

- Designed and deployed production-grade APIs in Golang for cloud-based call center platform
- Built core modules: Task Linking, RONA (Redirect On No Answer) configuration, and Label Management
- Integrated campaign services with RabbitMQ (producer/consumer model) for real-time data processing
- Created endpoints to process live URL calls and generate Excel-based reports for analytics and monitoring
- Ensured seamless microservices communication and deployed using Docker containers

## VoIP Development Expertise

- **SIP Protocol Mastery:** Deep understanding of SIP signaling (INVITE, ACK, BYE, CANCEL, REGISTER), call flow analysis, session management, and troubleshooting registration and authentication issues
- **RTP/RTCP Media Handling:** Practical experience with real-time media transmission, codec negotiation, quality monitoring (jitter, packet loss, latency), and one-way audio troubleshooting
- **FreeSWITCH ESL Integration:** Configured Event Socket Library connections, handled real-time call events (CHANNEL\_CREATE, ANSWER, BRIDGE, HANGUP, DESTROY), and built event-driven applications
- **Lua Scripting for Dialplans:** Developed custom Lua scripts for dynamic call routing, IVR systems, TTS/STT integrations using Google APIs, and call metadata capture
- **WebRTC Implementation:** Built browser-based communication using WebRTC with WebSocket signaling, SRTP secure media, and integration with SIP servers (FreeSWITCH)
- **Jitsi Architecture:** Understanding of Jitsi Meet components (Prosody XMPP server, Jicofo conference focus, JVB video bridge), SIP-to-WebRTC calling, and conferencing infrastructure
- **OpenSIPS Load Balancing:** Configured SIP routing, dispatcher module for load distribution, RTP Engine setup for media proxying, and NAT traversal
- **Docker VoIP Stack:** Containerized complex multi-component VoIP systems (FreeSWITCH, OpenSIPS, RTP Engine, MariaDB) using Docker Compose
- **Troubleshooting & Debugging:** Diagnosed ESL connection failures, ACL configuration errors, SIP registration issues, RTP audio problems, and analyzed call flows using logs and event tracing
- **Production Experience:** Hands-on deployment of production VoIP systems, performance optimization, call quality monitoring, and real-time system troubleshooting

## Additional Projects

### CRM Application

*Technologies:* Golang, JWT, MongoDB, Docker

- Built CRM backend in Golang with modular REST APIs and clean architecture principles
- Implemented JWT-based authentication (signup/login) with secure token management
- Integrated MongoDB for scalable document-based storage with optimized queries
- Containerized application using Docker for streamlined development and deployment

## Education

### **Bachelor of Computer Applications (BCA)**

Atmiya University – Rajkot, Gujarat

Graduation: 2025 | CGPA: 7.81

## Languages

- Hindi
- Gujarati
- English